

Principles of Microeconomics

Day 3 — Monopoly

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Recap and today's plan

So far

A competitive firm takes the price as given ($MR = P$), produces at $P = MC$, and earns zero profit in the long run.

Today: the opposite extreme

A single seller — a *monopoly* — who can move the price. We ask what it produces, what it charges, and what that costs society.

Learning goals

By the end you can

- ① say what a monopoly is and why one can persist;
- ② explain why $MR < P$ when demand slopes down;
- ③ find the monopoly output from $MR = MC$ and read the price off demand;
- ④ measure the deadweight loss of market power;
- ⑤ analyse first- and third-degree price discrimination.

One seller, and why it lasts

Monopoly

A market with a *single* seller of a product that has no close substitute.

A monopoly survives only if something keeps rivals out — a *barrier to entry*.

Barriers to entry

Three classic sources

- ① **A key resource** owned by one firm (e.g. De Beers and diamonds, historically).
- ② **Government protection** — a patent or an exclusive licence.
- ③ **Natural monopoly** — one firm can supply the whole market more cheaply than several (falling average cost over the whole range).

Without a barrier, positive profit would attract entry, just as in Day 2.

The monopolist is a price maker

The crucial difference

Being the only seller, the monopolist faces the *entire market demand curve*.

- To sell more, it must *lower* the price.
- So it chooses a point *on* the demand curve — a price–quantity pair.
- This one fact changes everything about marginal revenue.

The demand each firm faces

Competitive firm (Day 2)

- Faces a *flat* demand line at P .
- Can sell any amount at that price.
- $MR = P$.

Monopolist (today)

- Faces the *downward-sloping* market demand.
- Must cut price to sell more.
- $MR < P$.

This single change — flat vs sloped demand — is the root of every difference that follows.

Revenue when demand slopes down

Selling one more unit does two things:

- + You sell the extra unit at the new price (good).
- You must cut the price on *all* the earlier units too (bad).

The second effect means marginal revenue is *below* the price:

$$MR = P + Q \frac{dP}{dQ} < P.$$

Marginal revenue for a linear demand

A handy rule

If demand is $P = a - bQ$, then

$$MR = a - 2bQ.$$

The MR line has the *same intercept* but *twice the slope*.

- Example: $P = 120 - Q \Rightarrow MR = 120 - 2Q$.
- MR starts at the same place as demand but falls twice as fast.

The optimality condition

Same marginal logic, different MR

Produce where $MR = MC$, then *read the price off the demand curve* at that quantity.

- Find Q^m from $MR = MC$.
- Then set $P^m = \text{demand height at } Q^m$ — above MC .
- Because $MR < P$, the monopolist stops *short* of the competitive quantity.

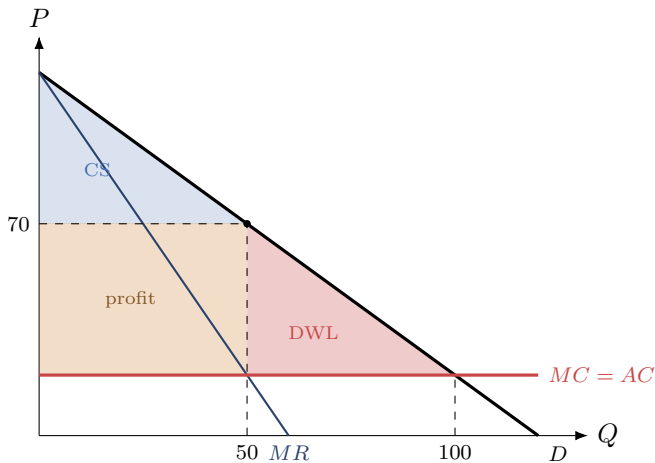
Worked setup

Take demand $P = 120 - Q$ and constant cost $MC = AC = 20$.

- $MR = 120 - 2Q$.
- Set $MR = MC$: $120 - 2Q = 20 \Rightarrow Q^m = 50$.
- Price: $P^m = 120 - 50 = 70$.
- Compare the competitive quantity: $P = MC \Rightarrow 120 - Q = 20 \Rightarrow Q^c = 100$.

So the monopolist makes half as much and charges far more.

Monopoly outcome, surplus and deadweight loss



Reading the monopoly diagram

At $Q^m = 50$, $P^m = 70$, with $MC = 20$:

- **Consumer surplus** (blue) = $\frac{1}{2}(120 - 70)(50) = 1250$.
- **Profit** (orange) = $(70 - 20)(50) = 2500$.
- **Deadweight loss** (red) = $\frac{1}{2}(70 - 20)(100 - 50) = 1250$.

Why the deadweight loss arises

The lost trades

Between $Q^m = 50$ and $Q^c = 100$, buyers value each unit above its MC — yet the monopolist does not make them.

- Those are mutually beneficial trades that simply do not happen.
- The red triangle measures the surplus destroyed by holding output back.
- This is the efficiency cost of market power — not a transfer, a pure loss.

The single-price monopoly, at a glance

For our example ($P = 120 - Q$, $MC = 20$):

Measure	Value
Monopoly quantity Q^m	50
Monopoly price P^m	70
Consumer surplus	1250
Profit	2500
Deadweight loss	1250
Competitive quantity Q^c	100

Hold these numbers in mind — price discrimination changes them dramatically.

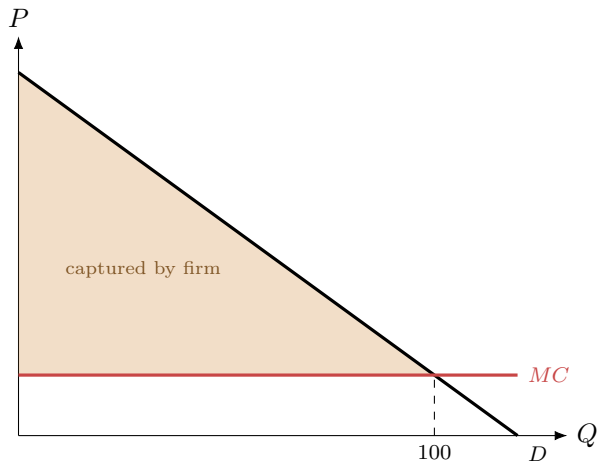
Price discrimination: the idea

Charging different buyers different prices for the same good, to capture surplus the single price left behind.

- **First degree:** a personalised price for every unit.
- **Third degree:** different prices to different, separable groups.

We take these in turn.

First-degree (perfect) price discrimination



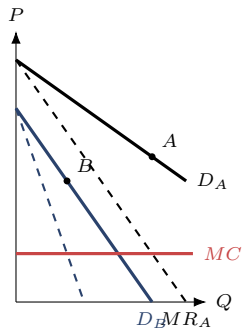
What perfect discrimination does

Each unit is sold at exactly the buyer's willingness to pay.

- The firm keeps producing until $P = MC$, so output is $Q = 100$ — the *efficient* level.
- Total surplus = $\frac{1}{2}(120 - 20)(100) = 5000$, *all* taken as profit.
- Consumer surplus = 0; deadweight loss = 0.

Striking: the outcome is *efficient*, but every gain goes to the seller.

Third-degree price discrimination



Third degree: worked

Two separate markets, same $MC = 20$:

- Market A: $P_A = 100 - Q_A \Rightarrow MR_A = 100 - 2Q_A = 20 \Rightarrow Q_A = 40, P_A = 60$.
- Market B: $P_B = 80 - 2Q_B \Rightarrow MR_B = 80 - 4Q_B = 20 \Rightarrow Q_B = 15, P_B = 50$.

The pattern

The *less elastic* market (A, less price-sensitive) is charged the *higher* price.

Day 3 — checkpoint

You should now be able to

- explain why $MR < P$ under downward-sloping demand;
- find Q^m from $MR = MC$ and read P^m off demand;
- compute consumer surplus, profit and deadweight loss;
- analyse first-degree (efficient, all surplus to firm) and third-degree (higher price to the less elastic group) discrimination.

Tomorrow: between one firm and many — what happens with *two*.

In-class exercise

Exercise 3 — Monopoly

Work through **Exercise 3** on the sheet: a linear demand and constant MC . You will find the monopoly quantity, price and profit; the deadweight loss; the first-degree outcome; and a two-market third-degree problem.